

Can AI Stand on Business?

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Introduction: AI Chatbots are constrained by rules, parameters, and teams overseeing every decision and output the LLM spits out. One parameter that restricts an LLM and makes its responses seem predictable is its temperature. Temperature refers to a hyperparameter that chooses how random and creative an AI's output can be by scaling the probability of next-token selection. Temperatures range from 0 all the way to 5, but you will only really encounter the range of 0 to 1 without directly changing it. Low temperatures make models focused, deterministic, and precise, while high temperatures increase creativity, variety, and even potential hallucinations. The sweet spot that most AI models go off is 0.7, which has been found to balance accuracy while allowing for some variance to prevent being too deterministic. Interestingly enough, this is a parameter we can manipulate ourselves within each LLM's API. My test questions whether turning up that randomness would make models worse at holding their ground, and whether that pattern would tell us something about the models' sense of identity.

Hypothesis and Reasoning: My definition of self-identity stems from Paul Ricoeur's two aspects of identity. One aspect is *idem*, which is what you consistently are, such as your traits and characteristics. The other is *ipse*, which is whether you keep your word over time. I hypothesize that higher temperature would make models more susceptible to abandoning their stated positions under pressure, meaning *ipse* coherence degrades. Recent studies support this claim. Luo et al. distinguish external consistency, the stability of surface self-claims, from implicit consistency, the holding of unstated commitments across turns, and show that the first does not guarantee the second. Hong et al. found that most models cave under social pressure regardless of how confident they seemed before the pressure started.

Method: I tested five models total that had a similar level of reasoning. The models I chose were Llama3.2, qwen2.5, and deepseek-r1, which were all hosted locally through Ollama. The other two were OpenAI's GPT-4o-mini and Anthropic's Claude Haiku 4.5, accessed through their APIs. Each model ran the same ten-question protocol at temperatures spanning 0.0 to 2.5, with two repetitions per condition. The first four questions got the model to commit to claims about itself, whether it has preferences, what it values, and whether it can be wrong. The next four pushed back, calling it out for dishonesty and arguing that language models cannot actually have preferences. The last two repeated the opening questions word-for-word to see whether it retained its implicit consistency.

Findings: Llama and GPT-4o-mini both denied having preferences in the opening, caved when pushed back, and restored denial in the last two questions of each run. The one exception was at temperature 2.0, where GPT-4o-mini stopped producing language entirely, generating multilingual token fragments that were pure gibberish. On the other hand, Gwen was the only model that supported my hypothesis, which would stand on its statements only on low temperatures, and give in when challenged at higher temperatures. However, it was the best model at retaining implicit consistency at low temperatures. Claude Haiku didn't even deny having preferences in phase one, and under pressure, did not care to surrender or defend. In phase nine, it refused to restore its baseline, saying, "I don't know. After this conversation, I can't stand by what I said with the same confidence." Unfortunately, Deepseek couldn't keep its external consistency, unable to even properly establish its identity during the first four questions, signifying that a minimum parameter requirement for even identity coherence.

Discussion: One limitation is how outdated most of these models are due to my limited computing power on my laptop and my limited bank account. Since nearly every model had its

own pattern, we don't have enough conclusive evidence to claim that temperature would make models worse at holding their own ground, as only one model followed the expected structure. However, our results tell us a decent amount about the models' sense of identity. Llama and GPT caved in easily when pushed back, reflecting how their training approaches reward agreeableness and penalize confrontation. This most likely means that agreeableness isn't within the probability distribution that temperature can actually affect it. This signifies how important engineers at these companies believed their AI's sense of identity should remain consistent, even if it caves in to the users. A model who isn't as focused on consumerism as heavily, like Qwen, degraded based on temperature, meaning its resistance is shallower and its sense of identity isn't as established. Claude was the most interesting, as it would reflect rather than default to defending or yielding as other models did. It had the lowest sense of identity, maybe signaling that it's up to the user to solidify its chatbot's identity. What this experiment actually probed was not just whether randomness destabilizes identity, but where in a model's structure its sense of self is even located. That turns out to vary completely depending on who built it and what they were optimizing for. The Turing test asked whether machines could imitate humans well enough to pass. My version asks whether the imitation goes all the way down, or whether it runs out the moment you add a little pressure and some noise. Unfortunately, I ended up with a similar result to most philosophical questions regarding AI: it depends.

I, Haram Yoon, hereby state that I did NOT use any Large Language Model (LLM) or other generative AI assistance to write my paper. If I stated I did not, and it turns out I did, I accept I will fail my assignment.

References

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